CORE PURE / DIFFERENTIAL EQUATIONS

Modelling with 1st Order Differential Equations

Worked solutions 14 questions · 225 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. A particle B, of mass 2 kg, moves in a straight line and has velocity $v\,\mathrm{m\,s^{-1}}$.

At time t seconds, where $t \geq 0$, a variable force of $\dfrac{12t-4v}{1+t}$ N acts on B.

There are no other forces acting on B. Initially, when $t = 0$, B has velocity $-5\,\mathrm{m\,s^{-1}}$.

The motion of B can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which B is stationary according to the model **[8]**

Solution

(a) The mass of B is 2 kg and the only force acting is

$$\frac{12t-4v}{1+t}$$

Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{12t-4v}{1+t}$$

Rearranging into the form

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

gives

$$\begin{aligned}\frac{\mathrm{d}v}{\mathrm{d}t} &= \frac{12t-4v}{2(1+t)} \\ &= \frac{6t-2v}{1+t}\end{aligned}$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{1+t}v = \frac{6t}{1+t}$$

Hence

$$P(t) = \frac{2}{1+t} \qquad \text{and} \qquad Q(t) = \frac{6t}{1+t}$$

(b) From part (a), the differential equation is

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{1+t}v = \frac{6t}{1+t}$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{2}{1+t}\,\mathrm{d}t} \\ &= e^{2\ln(1+t)} \\ &= (1+t)^2\end{aligned}$$

since $t \geq 0$, so $1 + t > 0$.

Multiplying the differential equation by $(1+t)^2$ gives

$$(1+t)^2\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2(1+t)^2}{1+t}v = \frac{6t(1+t)^2}{1+t}$$

$$(1+t)^2\frac{\mathrm{d}v}{\mathrm{d}t} + 2(1+t)v = 6t(1+t)$$

The left-hand side is the derivative of $(1+t)^2 v$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1+t)^2 v\big) = 6t(1+t)$$

Integrating with respect to t:

$$\begin{aligned}(1+t)^2 v &= \int 6t(1+t)\,\mathrm{d}t \\ &= \int (6t + 6t^2)\,\mathrm{d}t \\ &= 3t^2 + 2t^3 + c\end{aligned}$$

Use the initial condition $v = -5$ when $t = 0$:

$$\begin{aligned}(1+0)^2(-5) &= 3(0)^2 + 2(0)^3 + c \\ -5 &= c\end{aligned}$$

So

$$(1+t)^2 v = 3t^2 + 2t^3 - 5$$

and therefore

$$v = \frac{2t^3 + 3t^2 - 5}{(1+t)^2}$$

Factorising the numerator:

$$2t^3 + 3t^2 - 5 = (t-1)(2t^2 + 5t + 5)$$

so

$$v = \frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2}$$

The particle is stationary when $v = 0$, so we solve

$$\frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2} = 0$$

For $t \geq 0$, we have $(1+t)^2 > 0$, and also

$$2t^2 + 5t + 5 > 0$$

so the only way $v = 0$ is when

$$t - 1 = 0$$

Hence

$$t = 1$$

Therefore, the first time at which B is stationary is 1 s.

2. A tank initially contains 480 litres of water in which 300 grams of salt are dissolved.

At time $t = 0$, brine containing 4 grams of salt per litre begins to flow into the tank at a constant rate of 30 litres per minute. At the same time, the well-stirred mixture begins to be pumped out of the tank at a constant rate of 40 litres per minute.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid. There are x grams of salt in the tank t minutes after the flows begin.

(a) Find an expression for the volume of liquid in the tank at time t, and state the time at which the tank becomes empty. **[2]**

(b) Show that, while the tank still contains liquid,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 120 - \frac{4x}{48 - t}$$ **[3]**

(c) Hence find x in terms of t. **[4]**

(d) Find the time at which the amount of salt in the tank is greatest, and find this greatest amount. **[3]**

(e) When $t = 40$, a sample is taken and the tank is found to contain 290 grams of salt. Compare this measurement with the amount predicted by the model, and comment on the suitability of the model. **[2]**

Solution

(a) The volume changes at a rate of $30 - 40 = -10$ litres per minute, so the volume of liquid in the tank after t minutes is

$$V = 480 - 10t \text{ litres}$$

The tank is empty when $480 - 10t = 0$, that is, when $t = 48$ minutes.

(b) Salt enters at a rate of $4 \times 30 = 120$ grams per minute. The concentration in the tank is $\frac{x}{480-10t}$ grams per litre, so salt leaves at a rate of

$$40 \times \frac{x}{480 - 10t} = \frac{40x}{10(48 - t)} = \frac{4x}{48 - t} \text{ grams per minute}$$

The rate of change of the amount of salt is the rate in minus the rate out, so

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 120 - \frac{4x}{48 - t}$$

as required.

(c) Rearranging the equation from part (b):

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{4}{48 - t}x = 120$$

While the tank contains liquid, $48 - t > 0$. The integrating factor is

$$\mu(t) = e^{\int \frac{4}{48-t}\,\mathrm{d}t} = e^{-4\ln(48-t)} = (48 - t)^{-4}$$

Multiplying through by $(48-t)^{-4}$, the left-hand side is the derivative of $\mu(t)x = x(48-t)^{-4}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[x(48-t)^{-4}\right] = 120(48-t)^{-4}$$

$$x(48-t)^{-4} = \int 120(48-t)^{-4}\,\mathrm{d}t = 40(48-t)^{-3} + c$$

$$x = 40(48-t) + c(48-t)^4$$

When $t = 0$, $x = 300$:

$$300 = 40 \times 48 + c \times 48^4 \quad \Longrightarrow \quad c = -\frac{1620}{5\,308\,416} = -\frac{5}{16\,384}$$

Hence

$$x = 40(48-t) - \frac{5(48-t)^4}{16\,384}$$

(d) Differentiating the result of part (c):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -40 + \frac{20(48-t)^3}{16\,384}$$

Setting this equal to zero:

$$(48-t)^3 = \frac{40 \times 16\,384}{20} = 32\,768 \quad \Longrightarrow \quad 48 - t = 32 \quad \Longrightarrow \quad t = 16$$

Differentiating again,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{60(48-t)^2}{16\,384} < 0$$

so this is a maximum. When $t = 16$:

$$x = 40 \times 32 - \frac{5 \times 32^4}{16\,384} = 1280 - 320 = 960$$

The amount of salt is greatest 16 minutes after the flows begin, when the tank contains 960 grams of salt.

(e) When $t = 40$, $48 - t = 8$, so the model predicts

$$x = 40 \times 8 - \frac{5 \times 8^4}{16\,384} = 320 - 1.25 = 318.75$$

The model predicts about 319 grams, which is about 29 grams (roughly 10%) more than the 290 grams measured. The model is reasonably close, so it is fairly suitable, but it overestimates the salt; its assumption of instant, uniform mixing is less reliable when only 80 litres remain in the tank.

3. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are x grams of fertiliser in the tank after t minutes,

(a) show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 12 - \frac{3x}{200 - t} \qquad \textbf{[4]}$$

(b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. **[8]**

(c) Explain briefly one way in which the model could be refined. **[1]**

Solution

(a) Let V be the volume of liquid in the tank after t minutes.

Since liquid enters at $6\,\mathrm{L\,min^{-1}}$ and leaves at $9\,\mathrm{L\,min^{-1}}$, the volume is

$$\begin{aligned} V(t) &= 600 + (6 - 9)t \\ &= 600 - 3t \\ &= 3(200 - t) \end{aligned}$$

This is valid while the tank still contains liquid, so

$$600 - 3t > 0 \quad \Longrightarrow \quad t < 200$$

The fertiliser enters at

$$6 \times 2 = 12\,\mathrm{g\,min^{-1}}$$

The concentration in the tank at time t is

$$\frac{x}{V} = \frac{x}{600 - 3t}\,\mathrm{g\,L^{-1}}$$

So the fertiliser leaves at the rate

$$\begin{aligned} 9 \times \frac{x}{600 - 3t} &= \frac{9x}{600 - 3t} \\ &= \frac{9x}{3(200 - t)} \\ &= \frac{3x}{200 - t} \end{aligned}$$

Hence

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= \text{rate in} - \text{rate out} \\ &= 12 - \frac{3x}{200-t}\end{aligned}$$

So the model is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 12 - \frac{3x}{200-t}, \qquad t < 200$$

(b) From part (a):

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{200-t}x = 12$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{3}{200-t}\,\mathrm{d}t} \\ &= e^{-3\ln(200-t)} \\ &= (200-t)^{-3}\end{aligned}$$

Multiplying the differential equation by $(200-t)^{-3}$ gives

$$(200-t)^{-3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{200-t}(200-t)^{-3} = 12(200-t)^{-3}$$

The left-hand side is the derivative of $x(200-t)^{-3}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(x(200-t)^{-3}\big) = 12(200-t)^{-3}$$

Integrating:

$$\begin{aligned}x(200-t)^{-3} &= \int 12(200-t)^{-3}\,\mathrm{d}t \\ &= 6(200-t)^{-2} + c\end{aligned}$$

Therefore

$$\begin{aligned}x &= \big(6(200-t)^{-2} + c\big)(200-t)^3 \\ &= 6(200-t) + c(200-t)^3\end{aligned}$$

Initially the tank contains pure water, so $x = 0$ when $t = 0$. Substituting $t = 0$ gives

$$\begin{aligned}0 &= 6(200) + c(200)^3 \\ 0 &= 1200 + 8\,000\,000c \\ c &= -\frac{1200}{8\,000\,000} \\ &= -\frac{3}{20000}\end{aligned}$$

Hence

$$x = 6(200-t) - \frac{3}{20000}(200-t)^3$$

The volume at time t is

$$V = 3(200 - t)$$

So the concentration is

$$\frac{x}{V} = \frac{6(200-t) - \frac{3}{20000}(200-t)^3}{3(200-t)}$$

$$= 2 - \frac{(200-t)^2}{20000}$$

We want this concentration to be $1.5\,\mathrm{g\,L^{-1}}$, so

$$2 - \frac{(200-t)^2}{20000} = 1.5$$

$$\frac{(200-t)^2}{20000} = 0.5$$

$$(200-t)^2 = 10000$$

Since $t < 200$, we must have $200 - t > 0$. Therefore

$$200 - t = 100$$

So

$$t = 100$$

Therefore the concentration first reaches $1.5\,\mathrm{g\,L^{-1}}$ after 100 minutes.

(c) One refinement would be to remove the assumption of perfect instantaneous mixing, and allow the fertiliser concentration to vary in different parts of the tank.

4. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time t minutes after the flow begins, is modelled by $(4 + e^{-0.1t})$ mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let m be the amount of dye, in milligrams, in the tank at time t minutes after the flow begins.

(a) (i) Write down the volume of liquid in the tank at time t
(ii) Hence show that m satisfies the differential equation

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50} \qquad \textbf{[4]}$$

(b) By solving the differential equation, find an expression for m in terms of t **[5]**

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

(c) Assess the model in light of this information, giving a reason for your answer **[2]**

Solution

(a) (i) Convert the flow rates to litres per minute:

$$200\,\mathrm{ml\,min^{-1}} = 0.2\,\mathrm{L\,min^{-1}}, \qquad 100\,\mathrm{ml\,min^{-1}} = 0.1\,\mathrm{L\,min^{-1}}, \qquad 300\,\mathrm{ml\,min^{-1}} = 0.3\,\mathrm{L\,min^{-1}}$$

So the net rate of change of volume is

$$0.2 + 0.1 - 0.3 = 0\,\mathrm{L\,min^{-1}}$$

Hence the volume stays constant at its initial value of 15 litres:

$$V(t) = 15$$

So the volume of liquid in the tank at time t is 15 litres.

(ii) The rate at which dye enters through the first pipe is

$$0.2\big(4 + e^{-0.1t}\big)\,\mathrm{mg\,min^{-1}}$$

so

$$\text{rate in} = 0.8 + 0.2e^{-0.1t}$$

The second pipe contains pure water, so it contributes no dye.

Since the tank is well mixed and has volume 15 L, the concentration of dye in the tank is

$$\frac{m}{15}\,\mathrm{mg\,L^{-1}}$$

Therefore the rate at which dye leaves the tank is

$$0.3 \times \frac{m}{15} = \frac{0.3m}{15} = \frac{m}{50}\,\mathrm{mg\,min^{-1}}$$

Hence

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \text{rate in} - \text{rate out}$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$

as required.

(b) Rearranging the differential equation into linear form gives

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{1}{50}m = 0.8 + 0.2e^{-0.1t}$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{1}{50}\,\mathrm{d}t} \\ &= e^{t/50}\end{aligned}$$

Multiply the differential equation by $e^{t/50}$:

$$\begin{aligned}e^{t/50}\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{1}{50}e^{t/50}m &= 0.8e^{t/50} + 0.2e^{-0.1t}e^{t/50} \\ &= 0.8e^{t/50} + 0.2e^{-0.08t}\end{aligned}$$

The left-hand side is the derivative of $me^{t/50}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(me^{t/50}\right) = 0.8e^{t/50} + 0.2e^{-0.08t}$$

Integrating both sides with respect to t:

$$\begin{aligned}me^{t/50} &= \int 0.8e^{t/50}\,\mathrm{d}t + \int 0.2e^{-0.08t}\,\mathrm{d}t + c \\ &= 0.8(50)e^{t/50} + 0.2\left(\frac{1}{-0.08}\right)e^{-0.08t} + c \\ &= 40e^{t/50} - 2.5e^{-0.08t} + c\end{aligned}$$

Divide through by $e^{t/50}$:

$$\begin{aligned}m &= 40 - 2.5e^{-0.08t}e^{-t/50} + ce^{-t/50} \\ &= 40 - 2.5e^{-0.1t} + ce^{-0.02t}\end{aligned}$$

Initially there are 6 mg of dye in the tank, so $m(0) = 6$. Substituting $t = 0$ gives

$$\begin{aligned}6 &= 40 - 2.5 + c \\ 6 &= 37.5 + c \\ c &= -31.5\end{aligned}$$

Therefore

$$m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$$

(c) At $t = 30$ minutes, the model gives

$$m(30) = 40 - 2.5e^{-3} - 31.5e^{-0.6}$$

so the concentration is

$$\frac{m(30)}{15} = \frac{40 - 2.5e^{-3} - 31.5e^{-0.6}}{15} \approx 1.51\,\mathrm{mg\,L^{-1}}$$

The measured concentration was $1.52\,\mathrm{mg\,L^{-1}}$, which is very close to the model prediction. Therefore the model appears to be reasonable.

5. A particle B, of mass $2\,\text{kg}$, moves in a straight line and has velocity $v\,\text{m s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $-2t(v+3)\,\text{N}$ acts on B. There are no other forces acting on B. Initially, when $t = 0$, B has velocity $5\,\text{m s}^{-1}$.

The motion of B can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which B is stationary according to the model **[7]**

Solution

(a) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = -2t(v+3)$$

Expanding and dividing by 2 gives

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -tv - 3t$$

Hence

$$\frac{\mathrm{d}v}{\mathrm{d}t} + tv = -3t$$

Therefore, $P(t) = t$ and $Q(t) = -3t$.

(b) From part (a), the differential equation is

$$\frac{\mathrm{d}v}{\mathrm{d}t} + tv = -3t$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int t\,\mathrm{d}t} \\ &= e^{t^2/2}\end{aligned}$$

Multiplying the differential equation by $e^{t^2/2}$ gives

$$e^{t^2/2}\frac{\mathrm{d}v}{\mathrm{d}t} + tve^{t^2/2} = -3te^{t^2/2}$$

The left-hand side is the derivative of $ve^{t^2/2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t^2/2}\right) = -3te^{t^2/2}$$

Integrating with respect to t:

$$\begin{aligned} ve^{t^2/2} &= \int -3te^{t^2/2}\,\mathrm{d}t \\ &= -3e^{t^2/2} + c \end{aligned}$$

Therefore

$$v = -3 + ce^{-t^2/2}$$

Using the initial condition $v = 5$ when $t = 0$:

$$\begin{aligned} 5 &= -3 + ce^0 \\ 5 &= -3 + c \\ c &= 8 \end{aligned}$$

Hence

$$v = -3 + 8e^{-t^2/2}$$

For B to be stationary, $v = 0$. So

$$\begin{aligned} 0 &= -3 + 8e^{-t^2/2} \\ 8e^{-t^2/2} &= 3 \\ e^{-t^2/2} &= \frac{3}{8} \end{aligned}$$

Taking natural logarithms:

$$\begin{aligned} -\frac{t^2}{2} &= \ln\left(\frac{3}{8}\right) \\ \frac{t^2}{2} &= \ln\left(\frac{8}{3}\right) \\ t^2 &= 2\ln\left(\frac{8}{3}\right) \end{aligned}$$

Since $t \geq 0$, the first time is

$$t = \sqrt{2\ln\left(\frac{8}{3}\right)}$$

Therefore, the first time at which B is stationary is $\sqrt{2\ln\left(\frac{8}{3}\right)}\,\mathrm{s} \approx 1.40\,\mathrm{s}$.

6. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid, R litres per second, at time t seconds is measured from the instant the valve is opened. The cycle starts when $t = 0$ and there is initially no flow through the pipe. When $t = 1$, the flow rate is measured as $\pi e^{-1} \sin(1)$ litres per second.

While the cycle is occurring, R is modelled by the differential equation

$$\frac{\mathrm{d}R}{\mathrm{d}t} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

(a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t} \sin t \left(\arctan t + \frac{3\pi}{4} \right)$$ **[6]**

The cycle ends when there is again no flow through the pipe.

(b) Determine the length of time that the cycle lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form **[3]**

Solution

(a) For the cycle we consider $0 < t < \pi$, so $\sin t > 0$. The differential equation is

$$\frac{\mathrm{d}R}{\mathrm{d}t} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

The integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int (1 - \cot t)\, \mathrm{d}t} \\ &= e^{t - \int \cot t \, \mathrm{d}t} \\ &= e^{t - \ln(\sin t)} \\ &= \frac{e^t}{\sin t} \end{aligned}$$

Multiplying the differential equation by $\frac{e^t}{\sin t}$ gives

$$\begin{aligned} \frac{e^t}{\sin t} \frac{\mathrm{d}R}{\mathrm{d}t} + \frac{e^t}{\sin t}(1 - \cot t)R &= \frac{e^t}{\sin t} \times \frac{e^{-t} \sin t}{1 + t^2} \\ &= \frac{1}{1 + t^2} \end{aligned}$$

Now

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}\right) &= \frac{e^t}{\sin t} - \frac{e^t \cos t}{\sin^2 t} \\ &= \frac{e^t}{\sin t}\left(1 - \frac{\cos t}{\sin t}\right) \\ &= \frac{e^t}{\sin t}(1 - \cot t)\end{aligned}$$

The left-hand side is the derivative of $\frac{e^t}{\sin t}R$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}R\right) = \frac{1}{1+t^2}$$

Using the standard result

$$\int \frac{1}{t^2+a^2}\,\mathrm{d}t = \frac{1}{a}\arctan\left(\frac{t}{a}\right) + c$$

with $a = 1$, we get

$$\begin{aligned}\frac{e^t}{\sin t}R &= \int \frac{1}{1+t^2}\,\mathrm{d}t \\ &= \arctan t + c\end{aligned}$$

Therefore

$$R = e^{-t}\sin t(\arctan t + c)$$

Use the condition $R(1) = \pi e^{-1}\sin 1$:

$$\begin{aligned}\pi e^{-1}\sin 1 &= e^{-1}\sin 1(\arctan 1 + c) \\ &= e^{-1}\sin 1\left(\frac{\pi}{4} + c\right)\end{aligned}$$

Since $e^{-1}\sin 1 \neq 0$:

$$\begin{aligned}\pi &= \frac{\pi}{4} + c \\ c &= \frac{3\pi}{4}\end{aligned}$$

So

$$R = e^{-t}\sin t\left(\arctan t + \frac{3\pi}{4}\right)$$

as required.

(b) The cycle ends when there is again no flow, so we set $R = 0$:

$$e^{-t}\sin t\left(\arctan t + \frac{3\pi}{4}\right) = 0$$

For $t \geq 0$, we have $e^{-t} > 0$ and

$$\arctan t + \frac{3\pi}{4} > 0$$

so the only way $R = 0$ is if

$$\sin t = 0$$

Hence

$$t = n\pi \qquad (n \in \mathbb{Z})$$

and the first time after $t = 0$ is

$$t = \pi$$

Therefore the cycle lasts π seconds.

(c) The rate of increase of the flow rate at the start is the value of $\frac{\mathrm{d}R}{\mathrm{d}t}$ when $t = 0$.

From part (a):

$$R(t) = e^{-t} \sin t \left(\arctan t + \frac{3\pi}{4} \right)$$

Differentiate:

$$\frac{\mathrm{d}R}{\mathrm{d}t} = e^{-t} \left[\cos t \left(\arctan t + \frac{3\pi}{4} \right) + \frac{\sin t}{1 + t^2} - \sin t \left(\arctan t + \frac{3\pi}{4} \right) \right]$$

At the start of the cycle, when $t = 0$:

$$\frac{\mathrm{d}R}{\mathrm{d}t} = 1 \left[1 \times \frac{3\pi}{4} + 0 - 0 \right] = \frac{3\pi}{4}$$

So the rate of increase of the flow rate at the start is

$$\frac{3\pi}{4} \, \mathrm{L\,s^{-2}}$$

7. A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time t seconds is V litres. The pulse starts when $t = 0$, and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring, V is modelled by the differential equation

$$(3t - t^2)\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{(3t - t^2)^2}{1 + (t-1)^2} + (3 - 2t)V$$

(a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2)\left(\arctan(t-1) + 2\right)$$ **[6]**

The pulse ends when the tank is again empty.

(b) Determine the length of time that the pulse lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form **[3]**

Solution

(a) First rearrange the differential equation into linear form:

$$\frac{\mathrm{d}V}{\mathrm{d}t} - \frac{3 - 2t}{3t - t^2}V = \frac{3t - t^2}{1 + (t-1)^2}$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int -\frac{3-2t}{3t-t^2}\,\mathrm{d}t} \\ &= e^{-\ln(3t-t^2)} \\ &= \frac{1}{3t - t^2}\end{aligned}$$

since $\frac{\mathrm{d}}{\mathrm{d}t}(3t - t^2) = 3 - 2t$.

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{3t - t^2}\frac{\mathrm{d}V}{\mathrm{d}t} - \frac{3 - 2t}{(3t - t^2)^2}V = \frac{1}{1 + (t-1)^2}$$

The left-hand side is the derivative of $\frac{V}{3t-t^2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{V}{3t - t^2}\right) = \frac{1}{1 + (t-1)^2}$$

Using the standard result

$$\int \frac{1}{t^2 + a^2}\,\mathrm{d}t = \frac{1}{a}\arctan\left(\frac{t}{a}\right) + c$$

with $a = 1$ and t replaced by $t - 1$, we get

$$\frac{V}{3t - t^2} = \int \frac{1}{1 + (t-1)^2}\,\mathrm{d}t$$
$$= \arctan(t-1) + c$$

Use the given condition $V = 4$ when $t = 1$:

$$\frac{4}{3(1) - 1^2} = \arctan(1-1) + c$$
$$\frac{4}{2} = 0 + c$$
$$c = 2$$

Therefore

$$\frac{V}{3t - t^2} = \arctan(t-1) + 2$$

and so

$$V = (3t - t^2)(\arctan(t-1) + 2)$$

as required.

(b) The pulse ends when the tank is empty again, so set $V = 0$:

$$(3t - t^2)(\arctan(t-1) + 2) = 0$$

Now

$$-\frac{\pi}{2} < \arctan(t-1) < \frac{\pi}{2}$$

so

$$2 - \frac{\pi}{2} < \arctan(t-1) + 2 < 2 + \frac{\pi}{2}$$

Since $2 - \frac{\pi}{2} > 0$, the factor $\arctan(t-1) + 2$ is never zero. Hence

$$3t - t^2 = 0$$

so

$$t(3 - t) = 0$$
$$t = 0 \text{ or } t = 3$$

The pulse starts at $t = 0$, so it ends when $t = 3$. Therefore the pulse lasts 3 seconds.

(c) Differentiate the expression for V:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left[(3t - t^2)(\arctan(t-1) + 2)\right]$$
$$= (3 - 2t)(\arctan(t-1) + 2) + (3t - t^2)\left(\frac{1}{1 + (t-1)^2}\right)$$

At the start of the pulse, when $t = 0$:

$$\begin{aligned}\frac{\mathrm{d}V}{\mathrm{d}t} &= 3(\arctan(-1) + 2) + 0 \\ &= 3\left(-\frac{\pi}{4} + 2\right) \\ &= 6 - \frac{3\pi}{4}\end{aligned}$$

Therefore the rate of increase of volume at the start is $\left(6 - \frac{3\pi}{4}\right) \mathrm{L\,s^{-1}}$.

8. The concentration C, in suitable units, of a pollutant in a reservoir at time t hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is A, and its long-term value is 0.

(a) One simple model is to assume that the rate of decrease of concentration is directly proportional to C.

(i) Formulate a differential equation for this model **[1]**

(ii) Verify that $C = Ae^{-kt}$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition **[3]**

An alternative model uses the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = R(t)$$

where $R(t)$ is a function of t.

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2$$ **[2]**

(c) Suppose that $R(t) = 0$.

(i) Show that

$$C = \frac{A}{1+t^2}$$ **[4]**

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute **[2]**

(d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2}$$ **[6]**

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

(e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data **[2]**

Solution

(a) (i) If the rate of decrease of concentration is directly proportional to C, then

$$-\frac{\mathrm{d}C}{\mathrm{d}t} \propto C$$

so, for some constant $k > 0$:

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$$

(ii) Take

$$C = Ae^{-kt}$$

Then

$$\frac{\mathrm{d}C}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(Ae^{-kt}) = -kAe^{-kt}$$

Since $Ae^{-kt} = C$, this becomes

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$$

so the differential equation is satisfied.

For the initial condition:

$$C(0) = Ae^{-k\times 0} = A$$

so the initial concentration is correct.

For the long-term condition, since $k > 0$:

$$\lim_{t\to\infty} C = \lim_{t\to\infty} Ae^{-kt} = A \lim_{t\to\infty} e^{-kt} = 0$$

Hence $C = Ae^{-kt}$ satisfies the differential equation, the initial condition and the long-term condition.

(b) For

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = R(t)$$

the integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{2t}{1+t^2}\,\mathrm{d}t} \\ &= e^{\ln(1+t^2)} \\ &= 1 + t^2\end{aligned}$$

since $\frac{\mathrm{d}}{\mathrm{d}t}(1+t^2) = 2t$ and $1 + t^2 > 0$.

Hence the integrating factor is $1 + t^2$.

(c) (i) If $R(t) = 0$, the differential equation becomes

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = 0$$

Multiply through by the integrating factor $1 + t^2$:

$$(1+t^2)\frac{\mathrm{d}C}{\mathrm{d}t} + 2tC = 0$$

The left-hand side is the derivative of $(1 + t^2)C$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1+t^2)C\big) = 0$$

Integrating gives

$$(1+t^2)C = K$$

where K is a constant.

Use the initial condition $C(0) = A$:

$$(1 + 0^2)A = K$$

so $K = A$. Therefore

$$C = \frac{A}{1 + t^2}$$

(ii) Half the initial concentration is $\frac{A}{2}$, so set

$$\frac{A}{1 + t^2} = \frac{A}{2}$$

Cancelling A:

$$\frac{1}{1 + t^2} = \frac{1}{2}$$

so

$$1 + t^2 = 2$$

$$t^2 = 1$$

Since $t \geq 0$:

$$t = 1 \text{ hour}$$

Therefore the time is 60 minutes.

(d) If

$$R(t) = \frac{te^{-t}}{1 + t^2}$$

then the differential equation is

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1 + t^2}C = \frac{te^{-t}}{1 + t^2}$$

Multiply through by the integrating factor $1 + t^2$:

$$(1 + t^2)\frac{\mathrm{d}C}{\mathrm{d}t} + 2tC = te^{-t}$$

The left-hand side is the derivative of $(1 + t^2)C$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1 + t^2)C\big) = te^{-t}$$

Integrating:

$$(1 + t^2)C = \int te^{-t}\,\mathrm{d}t$$

Evaluate the integral by parts. The formula is

$$\int u\frac{\mathrm{d}v}{\mathrm{d}t}\,\mathrm{d}t = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t$$

Let

$$u = t, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = e^{-t}$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1, \qquad v = -e^{-t}$$

So

$$\begin{aligned}\int te^{-t}\,\mathrm{d}t &= uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t \\ &= t(-e^{-t}) - \int(-e^{-t})\,\mathrm{d}t \\ &= -te^{-t} + \int e^{-t}\,\mathrm{d}t \\ &= -te^{-t} - e^{-t} + K \\ &= -(1+t)e^{-t} + K\end{aligned}$$

Therefore

$$(1+t^2)C = -(1+t)e^{-t} + K$$

Use $C(0) = A$:

$$(1+0^2)A = -(1+0)e^{0} + K$$

$$A = -1 + K$$

so

$$K = A + 1$$

Hence

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2}$$

Also, using $\lim_{t\to\infty}(1+t)e^{-t} = 0$:

$$\lim_{t\to\infty} C = \lim_{t\to\infty} \frac{A + 1 - (1+t)e^{-t}}{1+t^2} = 0$$

so this model also has the correct long-term value.

(e) From part (c), model (c) predicts the concentration falls to half its initial value after 60 minutes. For model (d), with $A = 5$:

$$C = \frac{6 - (1+t)e^{-t}}{1+t^2}$$

Half the initial concentration is 2.5, so solve

$$\frac{6 - (1+t)e^{-t}}{1+t^2} = 2.5$$

Using a calculator gives

$$t \approx 1.055 \text{ hours}$$

which is

$$1.055 \times 60 \approx 63.3 \text{ minutes}$$

This is much closer to the observed 63 minutes than the 60 minutes from model (c). Therefore model (d) is more consistent with the data.

9. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time t minutes after pumping begins, is modelled by $2e^{-0.1t}$ mg per litre.

Let m be the amount of dye, in milligrams, in the vessel at time t minutes.

(a) Show that the volume of liquid in the vessel is $(4 + 0.1t)$ litres, and hence show that m satisfies

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \frac{2m}{40+t} \qquad \textbf{[4]}$$

(b) Solve this differential equation, given that $m = 4$ when $t = 0$, to find an expression for m in terms of t **[9]**

(c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer **[2]**

Solution

(a) The liquid enters at $300\,\mathrm{ml\,min^{-1}}$ and leaves at $200\,\mathrm{ml\,min^{-1}}$, so the net increase in volume is

$$300 - 200 = 100\,\mathrm{ml\,min^{-1}} = 0.1\,\mathrm{L\,min^{-1}}$$

Hence, starting from 4 L, the volume after t minutes is

$$V(t) = 4 + 0.1t$$

Now form a rate equation for the dye.

The dye enters with concentration $2e^{-0.1t}\,\mathrm{mg\,L^{-1}}$ and flow rate $0.3\,\mathrm{L\,min^{-1}}$, so the rate of dye entering is

$$0.3 \times 2e^{-0.1t} = 0.6e^{-0.1t}\,\mathrm{mg\,min^{-1}}$$

The concentration of dye in the vessel at time t is

$$\frac{m}{4+0.1t}\,\mathrm{mg\,L^{-1}}$$

Since liquid leaves at $0.2\,\mathrm{L\,min^{-1}}$, the rate of dye leaving is

$$0.2\left(\frac{m}{4+0.1t}\right)$$

Simplifying:

$$0.2\left(\frac{m}{4+0.1t}\right) = \frac{0.2m}{4+0.1t} = \frac{2m}{40+t}$$

Therefore

$$\begin{aligned}\frac{\mathrm{d}m}{\mathrm{d}t} &= \text{rate in} - \text{rate out} \\ &= 0.6e^{-0.1t} - \frac{2m}{40+t}\end{aligned}$$

So

$$V(t) = 4 + 0.1t \text{ litres}, \qquad \frac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \frac{2m}{40+t}$$

(b) We solve

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2m}{40+t} = 0.6e^{-0.1t}$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{2}{40+t}\,\mathrm{d}t} \\ &= e^{2\ln(40+t)} \\ &= (40+t)^2\end{aligned}$$

Multiplying the differential equation by $(40+t)^2$:

$$(40+t)^2\frac{\mathrm{d}m}{\mathrm{d}t} + (40+t)^2\frac{2m}{40+t} = 0.6(40+t)^2e^{-0.1t}$$

The left-hand side is the derivative of $(40+t)^2m$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((40+t)^2m\big) = 0.6(40+t)^2e^{-0.1t}$$

Integrating both sides:

$$(40+t)^2m = \int 0.6(40+t)^2e^{-0.1t}\,\mathrm{d}t$$

Expand $(40+t)^2$:

$$(40+t)^2 = t^2 + 80t + 1600$$

So

$$(40+t)^2m = \int 0.6(t^2+80t+1600)e^{-0.1t}\,\mathrm{d}t$$

Let

$$I = \int 0.6(t^2+80t+1600)e^{-0.1t}\,\mathrm{d}t$$

and use the integration by parts formula

$$\int u\frac{\mathrm{d}v}{\mathrm{d}t}\,\mathrm{d}t = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t$$

with

$$u = t^2 + 80t + 1600, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = 0.6e^{-0.1t}$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 2t + 80, \qquad v = \int 0.6e^{-0.1t}\,\mathrm{d}t = -6e^{-0.1t}$$

So

$$\begin{aligned} I &= uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 6\int (2t + 80)e^{-0.1t}\,\mathrm{d}t \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12\int (t + 40)e^{-0.1t}\,\mathrm{d}t \end{aligned}$$

Now let

$$J = \int (t + 40)e^{-0.1t}\,\mathrm{d}t$$

Again integrating by parts, take

$$u = t + 40, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = e^{-0.1t}$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1, \qquad v = \int e^{-0.1t}\,\mathrm{d}t = -10e^{-0.1t}$$

Hence

$$\begin{aligned} J &= uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t \\ &= -10(t + 40)e^{-0.1t} + 10\int e^{-0.1t}\,\mathrm{d}t \\ &= -10(t + 40)e^{-0.1t} - 100e^{-0.1t} + c_1 \\ &= -10(t + 50)e^{-0.1t} + c_1 \end{aligned}$$

Substituting this into the expression for I, and writing $c = 12c_1$:

$$\begin{aligned} I &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12\big(-10(t + 50)e^{-0.1t}\big) + c \\ &= -6e^{-0.1t}\big(t^2 + 80t + 1600 + 20t + 1000\big) + c \\ &= -6e^{-0.1t}(t^2 + 100t + 2600) + c \end{aligned}$$

Therefore

$$\int 0.6(t^2 + 80t + 1600)e^{-0.1t}\,\mathrm{d}t = -6e^{-0.1t}(t^2 + 100t + 2600) + c$$

where c is a constant.

So

$$(40 + t)^2 m = -6e^{-0.1t}(t^2 + 100t + 2600) + c$$

Now use the initial condition $m = 4$ when $t = 0$:

$$\begin{aligned}(40)^2(4) &= -6e^0(0^2 + 100(0) + 2600) + c \\ 6400 &= -6(2600) + c \\ 6400 &= -15600 + c\end{aligned}$$

Thus

$$c = 22000$$

Hence

$$(40 + t)^2 m = 22000 - 6e^{-0.1t}(t^2 + 100t + 2600)$$

and so

$$m = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40 + t)^2}$$

Therefore, the amount of dye is

$$m(t) = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40 + t)^2} \text{ mg}$$

(c) After 20 minutes, the model gives volume

$$V(20) = 4 + 0.1(20) = 6\,\text{L}$$

Using the expression for m:

$$\begin{aligned}m(20) &= \frac{22000 - 6e^{-2}(20^2 + 100(20) + 2600)}{(40 + 20)^2} \\ &= \frac{22000 - 6e^{-2}(400 + 2000 + 2600)}{60^2} \\ &= \frac{22000 - 30000e^{-2}}{3600} \\ &= \frac{55}{9} - \frac{25}{3}e^{-2}\end{aligned}$$

So the predicted concentration is

$$\begin{aligned}\frac{m(20)}{V(20)} &= \frac{1}{6}\left(\frac{55}{9} - \frac{25}{3}e^{-2}\right) \\ &= \frac{55}{54} - \frac{25}{18}e^{-2} \\ &\approx 0.831\,\text{mg}\,\text{L}^{-1}\end{aligned}$$

The measured concentration was $0.82\,\text{mg}\,\text{L}^{-1}$, which is very close to $0.831\,\text{mg}\,\text{L}^{-1}$.

So the model appears reasonable, since its prediction is consistent with the observed value.

10. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is m kg, and at time t seconds, for $0 \leq t \leq 10$, the boat's velocity is $v\,\mathrm{m\,s^{-1}}$.
A resistance to motion, modelled by a force of magnitude $0.2mv$ N, acts on the boat throughout the 10 seconds.

(a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant **[1]**

(b) During the drogue phase of the motion, for $5 \leq t \leq 10$, the drogue exerts an additional constant resistance force of magnitude m N and the engine provides no driving force. Show that, for $5 \leq t \leq 10$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$$ **[1]**

(c) (i) Solve the differential equation in part (b) **[5]**
(ii) Hence find the velocity of the boat when $t = 5$ **[1]**

(d) During the powered phase ($0 \leq t \leq 5$), the propeller provides a driving force of magnitude directly proportional to t^2. Show that, for $0 \leq t \leq 5$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2$$ **[1]**

where λ is a positive constant

(e) (i) Show by integration that, for $0 \leq t \leq 5$,

$$v = 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)$$ **[8]**

(ii) Hence find λ **[2]**

(f) Find the total distance, to the nearest metre, travelled by the boat during the motion **[6]**

Solution

(a) A resistive force depending on v is more realistic because water resistance increases as speed increases, and when $v = 0$ this model gives zero resistance. A constant resistive force would not reflect either of these features.

(b) During $5 \leq t \leq 10$, the only horizontal forces are the two resistances:

$$0.2mv \quad \text{and} \quad m$$

both acting opposite to the motion. Take the direction of motion as positive. Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -0.2mv - m$$

Dividing by m gives

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$$

as required.

(c) (i) For $5 \le t \le 10$, we solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}v = -1$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{1}{5}\,\mathrm{d}t} \\ &= e^{t/5}\end{aligned}$$

Multiplying the differential equation by $e^{t/5}$:

$$e^{t/5}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}e^{t/5}v = -e^{t/5}$$

The left-hand side is the derivative of $ve^{t/5}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) = -e^{t/5}$$

Integrating:

$$\begin{aligned}ve^{t/5} &= \int -e^{t/5}\,\mathrm{d}t \\ &= -5e^{t/5} + c\end{aligned}$$

Hence

$$v = -5 + ce^{-t/5}$$

The boat comes to rest at $t = 10$, so $v = 0$ when $t = 10$:

$$\begin{aligned}0 &= -5 + ce^{-2} \\ ce^{-2} &= 5 \\ c &= 5e^{2}\end{aligned}$$

Therefore

$$v = 5e^{2}e^{-t/5} - 5 = 5\left(e^{2-t/5} - 1\right)$$

So the solution is $v = 5\left(e^{2-t/5} - 1\right)$ for $5 \le t \le 10$.

(ii) Substituting $t = 5$ into the result from part (i):

$$\begin{aligned}v(5) &= 5\left(e^{2-1} - 1\right) \\ &= 5(e - 1)\end{aligned}$$

Hence the velocity when $t = 5$ is $5(e-1)\,\mathrm{m\,s^{-1}}$.

(d) Let the driving force during $0 \le t \le 5$ be $\lambda m t^2$, where $\lambda > 0$ is a constant. Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = \lambda m t^2 - 0.2mv$$

Dividing by m:

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2$$

as required.

(e) (i) We solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}v = \lambda t^2$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{1}{5}\,\mathrm{d}t} \\ &= e^{t/5}\end{aligned}$$

Multiplying through by $e^{t/5}$:

$$e^{t/5}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}e^{t/5}v = \lambda t^2 e^{t/5}$$

The left-hand side is the derivative of $ve^{t/5}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) = \lambda t^2 e^{t/5}$$

So

$$ve^{t/5} = \lambda \int t^2 e^{t/5}\,\mathrm{d}t + c$$

Let

$$I = \int t^2 e^{t/5}\,\mathrm{d}t$$

The formula for integration by parts is

$$\int u\frac{\mathrm{d}w}{\mathrm{d}t}\,\mathrm{d}t = uw - \int w\frac{\mathrm{d}u}{\mathrm{d}t}\,\mathrm{d}t$$

With $u = t^2$ and $\frac{\mathrm{d}w}{\mathrm{d}t} = e^{t/5}$, so that $\frac{\mathrm{d}u}{\mathrm{d}t} = 2t$ and $w = 5e^{t/5}$:

$$\begin{aligned}I &= 5t^2e^{t/5} - \int 10te^{t/5}\,\mathrm{d}t \\ &= 5t^2e^{t/5} - 10\int te^{t/5}\,\mathrm{d}t\end{aligned}$$

Now let

$$J = \int te^{t/5}\,\mathrm{d}t$$

Again integrating by parts, with $u = t$ and $\frac{\mathrm{d}w}{\mathrm{d}t} = e^{t/5}$, so that $\frac{\mathrm{d}u}{\mathrm{d}t} = 1$ and $w = 5e^{t/5}$:

$$\begin{aligned}J &= 5te^{t/5} - \int 5e^{t/5}\,\mathrm{d}t \\ &= 5te^{t/5} - 25e^{t/5}\end{aligned}$$

Hence

$$\begin{aligned} I &= 5t^2e^{t/5} - 10\Big(5te^{t/5} - 25e^{t/5}\Big) \\ &= 5t^2e^{t/5} - 50te^{t/5} + 250e^{t/5} \\ &= 5e^{t/5}\big(t^2 - 10t + 50\big) \end{aligned}$$

Therefore

$$\begin{aligned} ve^{t/5} &= \lambda\Big[5e^{t/5}\big(t^2 - 10t + 50\big)\Big] + c \\ v &= 5\lambda\big(t^2 - 10t + 50\big) + ce^{-t/5} \end{aligned}$$

The boat starts from rest, so $v = 0$ when $t = 0$:

$$\begin{aligned} 0 &= 5\lambda(50) + c \\ c &= -250\lambda \end{aligned}$$

Hence

$$\begin{aligned} v &= 5\lambda\big(t^2 - 10t + 50\big) - 250\lambda e^{-t/5} \\ &= 5\lambda\Big(t^2 - 10t + 50 - 50e^{-t/5}\Big) \end{aligned}$$

Thus, for $0 \leq t \leq 5$:

$$v = 5\lambda\Big(t^2 - 10t + 50 - 50e^{-t/5}\Big)$$

(ii) At $t = 5$, the expressions for v from parts (c)(ii) and (e)(i) must be equal.

From part (e)(i):

$$\begin{aligned} v(5) &= 5\lambda\big(25 - 50 + 50 - 50e^{-1}\big) \\ &= 5\lambda\left(25 - \frac{50}{e}\right) \end{aligned}$$

From part (c)(ii):

$$v(5) = 5(e - 1)$$

Equating these:

$$\begin{aligned} 5\lambda\left(25 - \frac{50}{e}\right) &= 5(e - 1) \\ \lambda\left(25 - \frac{50}{e}\right) &= e - 1 \\ \lambda\left(\frac{25e - 50}{e}\right) &= e - 1 \\ \lambda\left(\frac{25(e - 2)}{e}\right) &= e - 1 \end{aligned}$$

So

$$\lambda = \frac{e(e - 1)}{25(e - 2)}$$

Therefore

$$\lambda = \frac{e(e-1)}{25(e-2)} \approx 0.260$$

(f) Since the boat moves forwards throughout the motion, the total distance is

$$s = \int_0^{10} v \, \mathrm{d}t = \int_0^5 v \, \mathrm{d}t + \int_5^{10} v \, \mathrm{d}t$$

For the powered phase:

$$\begin{aligned} s_1 &= \int_0^5 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right) \mathrm{d}t \\ &= 5\lambda\left[\frac{t^3}{3} - 5t^2 + 50t + 250e^{-t/5}\right]_0^5 \\ &= 5\lambda\left(\frac{125}{3} - 125 + 250 + \frac{250}{e} - 250\right) \\ &= 5\lambda\left(\frac{125}{3} - 125 + \frac{250}{e}\right) \\ &= 5\lambda\left(-\frac{250}{3} + \frac{250}{e}\right) \\ &= 1250\lambda\left(\frac{3-e}{3e}\right) \end{aligned}$$

Substituting $\lambda = \frac{e(e-1)}{25(e-2)}$:

$$\begin{aligned} s_1 &= 1250\left(\frac{e(e-1)}{25(e-2)}\right)\left(\frac{3-e}{3e}\right) \\ &= \frac{50(e-1)(3-e)}{3(e-2)} \end{aligned}$$

So

$$s_1 \approx 11.2\,\mathrm{m}$$

For the drogue phase:

$$\begin{aligned} s_2 &= \int_5^{10} 5\left(e^{2-t/5} - 1\right) \mathrm{d}t \\ &= 5\left[\int_5^{10} e^{2-t/5} \, \mathrm{d}t - \int_5^{10} 1 \, \mathrm{d}t\right] \\ &= 5\left[-5e^{2-t/5} - t\right]_5^{10} \\ &= 5((-5-10) - (-5e-5)) \\ &= 25(e-2) \end{aligned}$$

So

$$s_2 \approx 18.0\,\mathrm{m}$$

Therefore the total distance is

$$\begin{aligned} s &= s_1 + s_2 \\ &= \frac{50(e-1)(3-e)}{3(e-2)} + 25(e-2) \\ &\approx 11.2 + 18.0 \\ &\approx 29.2 \end{aligned}$$

Hence the total distance travelled, to the nearest metre, is 29 m.

11. A winch raises a lift cage vertically, starting from rest at time $t = 0$. The combined mass of the cage and its load is m kg, and at time t seconds the cage has upward velocity $v\,\mathrm{m\,s^{-1}}$.
A resistance to the motion, modelled by a force of magnitude $0.5mv$ N, acts on the cage while it is rising.
In this question, take $g = 9.8\,\mathrm{m\,s^{-2}}$.
During the first stage of the motion, the winch provides an upward tension of magnitude $m\big(g+6e^{-t/4}\big)$ N.

(a) Show that, during the first stage,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.5v = 6e^{-t/4}$$

[2]

(b) Hence show that, during the first stage,

$$v = 24\big(e^{-t/4} - e^{-t/2}\big)$$

[5]

(c) The first stage ends at the instant when the speed of the cage is greatest.

(i) Show that the first stage ends when $t = 4\ln 2$, and find the speed of the cage at this instant. **[3]**

(ii) Find the distance through which the cage rises during the first stage. **[3]**

(d) During the second stage, for $t \geq 4\ln 2$, the winch provides a constant upward tension of magnitude $m(g-2)$ N. The second stage ends when the cage first comes to rest.

(i) Find an expression for v in terms of t during the second stage. **[4]**

(ii) Show that the cage comes to rest when $t = \ln 100$. **[2]**

(iii) Find the total distance through which the cage rises, giving your answer to 3 significant figures. **[4]**

(e) Explain why the winch must change its tension at the instant $t = \ln 100$ if the cage is to remain at rest. **[1]**

Solution

(a) The forces on the cage are the tension upwards, and the weight mg and the resistance $0.5mv$ downwards.

Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = m\big(g + 6e^{-t/4}\big) - mg - 0.5mv$$

Dividing by m and rearranging gives

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.5v = 6e^{-t/4}$$

as required.

(b) The integrating factor is

$$\mu(t) = e^{\int 0.5\,\mathrm{d}t} = e^{t/2}$$

Multiplying the equation from part (a) by $e^{t/2}$:

$$e^{t/2}\frac{\mathrm{d}v}{\mathrm{d}t} + 0.5e^{t/2}v = 6e^{t/4}$$

The left-hand side is the derivative of $\mu(t)v$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/2}\right) = 6e^{t/4}$$

$$ve^{t/2} = \int 6e^{t/4}\,\mathrm{d}t = 24e^{t/4} + c$$

The cage starts from rest, so $v = 0$ when $t = 0$:

$$0 = 24 + c \quad \Longrightarrow \quad c = -24$$

Hence

$$ve^{t/2} = 24e^{t/4} - 24 \quad \Longrightarrow \quad v = 24\left(e^{-t/4} - e^{-t/2}\right)$$

as required.

(c) (i) The speed is greatest where the acceleration is zero. Differentiating the result of part (b):

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -6e^{-t/4} + 12e^{-t/2}$$

Setting this equal to zero:

$$12e^{-t/2} = 6e^{-t/4} \quad \Longrightarrow \quad e^{t/4} = 2 \quad \Longrightarrow \quad t = 4\ln 2$$

At this instant $e^{-t/4} = \frac{1}{2}$ and $e^{-t/2} = \frac{1}{4}$, so

$$v = 24\left(\frac{1}{2} - \frac{1}{4}\right) = 6$$

The speed of the cage is $6\,\mathrm{m\,s^{-1}}$.

(ii) The cage moves upwards throughout the first stage, so the distance is

$$\begin{aligned}\int_0^{4\ln 2} 24\left(e^{-t/4} - e^{-t/2}\right)\mathrm{d}t &= 24\left[-4e^{-t/4} + 2e^{-t/2}\right]_0^{4\ln 2} \\ &= 24\left(\left(-2 + \frac{1}{2}\right) - (-4 + 2)\right) \\ &= 24 \times \frac{1}{2} = 12\end{aligned}$$

The cage rises 12 m during the first stage.

(d) (i) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = m(g - 2) - mg - 0.5mv \quad \Longrightarrow \quad \frac{\mathrm{d}v}{\mathrm{d}t} + 0.5v = -2$$

The integrating factor is again $\mu(t) = e^{t/2}$. The left-hand side, multiplied by $e^{t/2}$, is the derivative of $\mu(t)v$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/2}\right) = -2e^{t/2}$$

$$ve^{t/2} = -4e^{t/2} + c$$

$$v = -4 + ce^{-t/2}$$

The velocity is continuous at the change of tension, so $v = 6$ when $t = 4\ln 2$, where $e^{-t/2} = \frac{1}{4}$:

$$6 = -4 + \frac{c}{4} \quad \Longrightarrow \quad c = 40$$

Hence, during the second stage,

$$v = 40e^{-t/2} - 4$$

(ii) Setting $v = 0$:

$$40e^{-t/2} = 4 \quad \Longrightarrow \quad e^{t/2} = 10 \quad \Longrightarrow \quad t = 2\ln 10 = \ln 100$$

as required.

(iii) The distance risen during the second stage is

$$\begin{aligned}\int_{4\ln 2}^{\ln 100} \left(40e^{-t/2} - 4\right) \mathrm{d}t &= \left[-80e^{-t/2} - 4t\right]_{4\ln 2}^{\ln 100} \\ &= (-8 - 4\ln 100) - (-20 - 16\ln 2) \\ &= 12 - 8\ln 10 + 16\ln 2 \\ &= 12 - 8\ln\frac{5}{2}\end{aligned}$$

Adding the 12 m from part (c)(ii), the total distance is

$$24 - 8\ln\frac{5}{2} \approx 16.7\,\mathrm{m} \qquad (3\text{ s.f.})$$

(e) At rest there is no resistance, so the resultant upward force on the cage is $m(g-2) - mg = -2m$, which is downwards: with the same tension the cage would accelerate downwards, so the tension must increase to mg N to hold it at rest.

12. A tank initially contains 10 litres of solution in which 20 grams of dye are dissolved.
A dye solution containing 6 grams of dye per litre is pumped into the tank at a constant rate of 4 litres per minute and mixes instantly and uniformly with the liquid already in the tank.
At the same time, liquid is removed from the tank at a constant rate of 3 litres per minute.
After t minutes, the tank contains m grams of dye.

(a) Show that

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{3m}{10+t} = 24 \qquad \textbf{[4]}$$

(b) Solve this differential equation to find m in terms of t. **[6]**

(c) (i) Find the concentration of dye in the tank, in grams per litre, in terms of t, and explain why it increases as t increases. **[3]**

(ii) State the limiting value of the concentration predicted by the model. **[1]**

(d) Find the time at which the concentration of dye in the tank first reaches 5.75 grams per litre. **[3]**

(e) The tank has a capacity of 50 litres.
Find the concentration of dye in the tank at the moment it becomes full, and comment on the limiting value found in part (c)(ii). **[3]**

Solution

(a) The volume increases at $4 - 3 = 1$ litre per minute, so after t minutes the volume of liquid in the tank is $(10 + t)$ litres, and the concentration of dye in the tank is $\frac{m}{10+t}$ grams per litre.

Dye enters at $4 \times 6 = 24$ grams per minute. Liquid leaves at 3 litres per minute with concentration $\frac{m}{10+t}$, so dye leaves at $\frac{3m}{10+t}$ grams per minute. Hence

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 24 - \frac{3m}{10+t}$$

which rearranges to

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{3m}{10+t} = 24$$

as required.

(b) The integrating factor is

$$\mu(t) = e^{\int \frac{3}{10+t}\,\mathrm{d}t} = e^{3\ln(10+t)} = (10+t)^3$$

since $10 + t > 0$. Multiplying the equation by $(10 + t)^3$:

$$(10+t)^3\frac{\mathrm{d}m}{\mathrm{d}t} + 3(10+t)^2 m = 24(10+t)^3$$

The left-hand side is the derivative of $\mu(t)m = (10 + t)^3 m$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[(10+t)^3 m\right] = 24(10+t)^3$$

$$(10+t)^3 m = 6(10+t)^4 + c$$

When $t = 0$, $m = 20$:

$$1000 \times 20 = 6 \times 10\,000 + c \quad \Longrightarrow \quad c = -40\,000$$

Dividing by $(10 + t)^3$:

$$m = 6(10 + t) - \frac{40\,000}{(10 + t)^3}$$

(c) (i) The concentration, in grams per litre, is

$$\frac{m}{10 + t} = 6 - \frac{40\,000}{(10 + t)^4}$$

As t increases, $(10 + t)^4$ increases, so $\frac{40\,000}{(10+t)^4}$ decreases. A smaller amount is subtracted from 6, so the concentration increases as t increases.

(ii) As $t \to \infty$, $\frac{40\,000}{(10+t)^4} \to 0$, so the concentration tends to 6 grams per litre.

(d) Setting the concentration equal to 5.75:

$$6 - \frac{40\,000}{(10 + t)^4} = 5.75$$

$$(10 + t)^4 = 160\,000$$

$$10 + t = 20$$

taking the positive root since $10 + t > 0$. The concentration increases with t, so it first reaches 5.75 grams per litre when $t = 10$, after 10 minutes.

(e) The tank is full when $10 + t = 50$, that is, when $t = 40$. The concentration is then

$$6 - \frac{40\,000}{50^4} = 6 - 0.0064 = 5.9936$$

so the concentration is 5.99 grams per litre (3 s.f.). After $t = 40$ the tank overflows, so the model no longer applies. The concentration therefore never reaches the limiting value of 6 grams per litre found in part (c)(ii); it reaches only about 5.99 grams per litre.

13. A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of p litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of q litres per minute.

After t minutes, let x litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

(a) (i) Show that the proportion of fruit concentrate in the tank after t minutes is

$$\frac{x}{4+(p-q)t}$$ **[2]**

(ii) Hence show that

$$\frac{\mathrm{d}x}{\mathrm{d}t}+\frac{qx}{4+(p-q)t}=\frac{3p}{5}$$ **[2]**

(b) Now consider the case where $q=p$.

(i) Solve the differential equation to find x in terms of p and t. **[5]**

(ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of p. **[2]**

(c) Now consider the case where $q=2p$.

(i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{4}{p}$. **[1]**

(ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. **[10]**

Solution

(a) (i) After t minutes, the volume in the tank is

$$4+pt-qt=4+(p-q)t$$

Since there are x litres of fruit concentrate in the tank, the proportion of fruit concentrate is

$$\frac{x}{4+(p-q)t}$$

(ii) The flavouring liquid entering the tank is 60% concentrate, so the rate at which concentrate enters is

$$0.6p=\frac{3p}{5}$$

From part (i), the proportion of concentrate in the tank is $\frac{x}{4+(p-q)t}$, so the rate at which concentrate leaves is

$$q\left(\frac{x}{4+(p-q)t}\right)=\frac{qx}{4+(p-q)t}$$

Therefore

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= \text{rate in} - \text{rate out} \\ &= \frac{3p}{5} - \frac{qx}{4+(p-q)t}\end{aligned}$$

Rearranging:

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{qx}{4+(p-q)t} = \frac{3p}{5}$$

as required.

(b) (i) If $q = p$, then the differential equation from part (a)(ii) becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{px}{4+(p-p)t} = \frac{3p}{5}$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{p}{4}x = \frac{3p}{5}$$

Initially the tank contains only water, so

$$x(0) = 0$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{p}{4}\,\mathrm{d}t} \\ &= e^{pt/4}\end{aligned}$$

Multiplying the differential equation by $e^{pt/4}$ gives

$$e^{pt/4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{p}{4}e^{pt/4}x = \frac{3p}{5}e^{pt/4}$$

The left-hand side is the derivative of $xe^{pt/4}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(xe^{pt/4}\right) = \frac{3p}{5}e^{pt/4}$$

Integrating:

$$\begin{aligned}xe^{pt/4} &= \int \frac{3p}{5}e^{pt/4}\,\mathrm{d}t \\ &= \frac{3p}{5} \times \frac{4}{p}e^{pt/4} + c \\ &= \frac{12}{5}e^{pt/4} + c\end{aligned}$$

Hence

$$x = \frac{12}{5} + ce^{-pt/4}$$

Using $x(0) = 0$:

$$0 = \frac{12}{5} + c$$

so

$$c = -\frac{12}{5}$$

Therefore

$$x = \frac{12}{5}\left(1 - e^{-pt/4}\right)$$

(ii) When $q = p$, the total volume stays constant at 4 litres.

After 4 minutes, the liquid is 30% concentrate, so the amount of concentrate is

$$x = 0.3 \times 4 = \frac{6}{5}$$

Substitute $t = 4$ into the formula from part (i):

$$\frac{12}{5}\left(1 - e^{-p}\right) = \frac{6}{5}$$

Multiply through by 5:

$$12(1 - e^{-p}) = 6$$

Hence

$$1 - e^{-p} = \frac{1}{2}$$

$$e^{-p} = \frac{1}{2}$$

Taking natural logarithms:

$$-p = \ln\left(\frac{1}{2}\right) = -\ln 2$$

so

$$p = \ln 2$$

Therefore $p = \ln 2 \approx 0.693\,\mathrm{L\,min^{-1}}$.

(c) (i) If $q = 2p$, then the volume after t minutes is

$$4 + (p - 2p)t = 4 - pt$$

At

$$t = \frac{4}{p}$$

this volume is 0, so the tank is empty. For $t \geq \frac{4}{p}$, the denominator $4 - pt$ is zero or negative, which is not physically meaningful.

Therefore the differential equation is only valid for $0 \leq t < \frac{4}{p}$.

(ii) With $q = 2p$, the differential equation becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2px}{4 - pt} = \frac{3p}{5}$$

for $0 \leq t < \frac{4}{p}$, with

$$x(0) = 0$$

The integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int \frac{2p}{4-pt}\,\mathrm{d}t} \\ &= e^{-2\ln(4-pt)} \\ &= (4-pt)^{-2}\end{aligned}$$

since $\frac{\mathrm{d}}{\mathrm{d}t}(4-pt) = -p$ and $4 - pt > 0$.

Multiply the differential equation by $(4-pt)^{-2}$:

$$(4-pt)^{-2}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2p}{4-pt}(4-pt)^{-2}x = \frac{3p}{5}(4-pt)^{-2}$$

The left-hand side is the derivative of $x(4-pt)^{-2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(x(4-pt)^{-2}\big) = \frac{3p}{5}(4-pt)^{-2}$$

Integrating:

$$\begin{aligned}x(4-pt)^{-2} &= \int \frac{3p}{5}(4-pt)^{-2}\,\mathrm{d}t \\ &= \frac{3}{5(4-pt)} + c\end{aligned}$$

Multiply by $(4-pt)^2$:

$$x = \frac{3}{5}(4-pt) + c(4-pt)^2$$

Use $x(0) = 0$:

$$\begin{aligned}0 &= \frac{3}{5}(4) + c(4)^2 \\ 0 &= \frac{12}{5} + 16c\end{aligned}$$

so

$$c = -\frac{3}{20}$$

Hence

$$\begin{aligned}x &= \frac{3}{5}(4-pt) - \frac{3}{20}(4-pt)^2 \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{3}{20}\big(16 - 8pt + p^2t^2\big) \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{12}{5} + \frac{6pt}{5} - \frac{3p^2t^2}{20} \\ &= \frac{3pt}{5} - \frac{3p^2t^2}{20}\end{aligned}$$

So

$$x = \frac{3pt}{5} - \frac{3p^2t^2}{20} = \frac{3}{20}pt(4 - pt)$$

To find the maximum value, differentiate:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3p}{5} - \frac{3p^2t}{10}$$

Set this equal to 0:

$$\frac{3p}{5} - \frac{3p^2t}{10} = 0$$

$$\frac{3p}{10}(2 - pt) = 0$$

Since $p > 0$, this gives

$$2 - pt = 0$$

so

$$t = \frac{2}{p}$$

This lies in the interval $0 \leq t < \frac{4}{p}$, and since the quadratic has negative coefficient of t^2, this gives the maximum.

Now substitute $t = \frac{2}{p}$:

$$\begin{aligned} x_{\max} &= \frac{3p}{5}\left(\frac{2}{p}\right) - \frac{3p^2}{20}\left(\frac{2}{p}\right)^2 \\ &= \frac{6}{5} - \frac{3}{20} \times 4 \\ &= \frac{6}{5} - \frac{12}{20} \\ &= \frac{6}{5} - \frac{3}{5} \\ &= \frac{3}{5} \end{aligned}$$

Therefore the maximum amount of fruit concentrate is $\frac{3}{5}$ litres, occurring at $t = \frac{2}{p}$ minutes.

14. The concentration C, in suitable units, of a drug in a patient's bloodstream at time t hours is modelled by the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{t+5}{(t+1)(t+3)}C = S(t), \qquad t \geq 0$$

where $S(t)$ represents the rate at which the drug enters the bloodstream.

(a) Show that an integrating factor for this differential equation is

$$\frac{(t+1)^2}{t+3} \qquad \textbf{[5]}$$

In one treatment, a single injection at time $t = 0$ gives an initial concentration of 9, and no more of the drug then enters the bloodstream, so that $S(t) = 0$.

(b) Show that, for this treatment,

$$C = \frac{3(t+3)}{(t+1)^2} \qquad \textbf{[4]}$$

In a second treatment, the drug is absorbed gradually from a tablet taken at time $t = 0$, so that $C = 0$ when $t = 0$ and $S(t) = \frac{24}{(t+1)^2(t+3)}$ for $t \geq 0$.

(c) (i) Show that, for this treatment,

$$C = \frac{8t}{(t+1)^2} \qquad \textbf{[4]}$$

(ii) Find the greatest concentration predicted for this treatment, and the time at which it occurs. **[3]**

(d) Find the time, in hours and minutes, at which the two treatments give the same concentration. **[2]**

(e) Show that both treatments predict that the concentration tends to zero in the long term, and state, with a reason, which treatment gives the greater concentration for large values of t. **[2]**

Solution

(a) Write the coefficient of C in partial fractions:

$$\frac{t+5}{(t+1)(t+3)} \equiv \frac{A}{t+1} + \frac{B}{t+3} \quad \Longrightarrow \quad t+5 \equiv A(t+3) + B(t+1)$$

Putting $t = -1$ gives $4 = 2A$, so $A = 2$. Putting $t = -3$ gives $2 = -2B$, so $B = -1$. Hence

$$\frac{t+5}{(t+1)(t+3)} = \frac{2}{t+1} - \frac{1}{t+3}$$

For $t \geq 0$, both $t+1$ and $t+3$ are positive, so

$$\int \frac{t+5}{(t+1)(t+3)}\,\mathrm{d}t = 2\ln(t+1) - \ln(t+3) = \ln\left(\frac{(t+1)^2}{t+3}\right)$$

The integrating factor is

$$\mu(t) = e^{\int \frac{t+5}{(t+1)(t+3)}\,\mathrm{d}t} = e^{\ln\left(\frac{(t+1)^2}{t+3}\right)} = \frac{(t+1)^2}{t+3}$$

as required.

(b) With $S(t) = 0$, multiplying the differential equation by $\mu(t)$ gives

$$\frac{(t+1)^2}{t+3}\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{(t+1)^2}{t+3} \times \frac{t+5}{(t+1)(t+3)}C = 0$$

The left-hand side is the derivative of $\mu(t)C$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{(t+1)^2}{t+3}C\right) = 0$$

$$\frac{(t+1)^2}{t+3}C = K$$

When $t = 0$, $C = 9$:

$$\frac{1}{3} \times 9 = K \quad \Longrightarrow \quad K = 3$$

Therefore

$$C = \frac{3(t+3)}{(t+1)^2}$$

as required.

(c) (i) Multiplying the differential equation by $\mu(t)$, the left-hand side is the derivative of $\mu(t)C$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{(t+1)^2}{t+3}C\right) = \frac{(t+1)^2}{t+3} \times \frac{24}{(t+1)^2(t+3)} = \frac{24}{(t+3)^2}$$

Integrating both sides:

$$\frac{(t+1)^2}{t+3}C = -\frac{24}{t+3} + K$$

When $t = 0$, $C = 0$:

$$0 = -8 + K \quad \Longrightarrow \quad K = 8$$

so

$$\frac{(t+1)^2}{t+3}C = 8 - \frac{24}{t+3} = \frac{8(t+3)-24}{t+3} = \frac{8t}{t+3}$$

$$C = \frac{8t}{(t+1)^2}$$

as required.

(ii) Using the quotient rule:

$$\frac{\mathrm{d}C}{\mathrm{d}t} = \frac{8(t+1)^2 - 8t \times 2(t+1)}{(t+1)^4} = \frac{8(t+1)-16t}{(t+1)^3} = \frac{8(1-t)}{(t+1)^3}$$

This is zero when $t = 1$. For $0 \leq t < 1$, $\frac{\mathrm{d}C}{\mathrm{d}t} > 0$, and for $t > 1$, $\frac{\mathrm{d}C}{\mathrm{d}t} < 0$, so this is the greatest concentration:

$$C = \frac{8 \times 1}{2^2} = 2$$

The greatest concentration is 2, and it occurs 1 hour after the tablet is taken.

(d) Using part (b) and part (c)(i), the concentrations are equal when

$$\frac{3(t+3)}{(t+1)^2} = \frac{8t}{(t+1)^2}$$

$$3t + 9 = 8t$$

$$t = \frac{9}{5} = 1.8$$

The two treatments give the same concentration after 1 hour 48 minutes.

(e) Dividing the numerator and denominator of each expression by t^2:

$$\frac{3(t+3)}{(t+1)^2} = \frac{\frac{3}{t} + \frac{9}{t^2}}{\left(1 + \frac{1}{t}\right)^2} \to 0, \qquad \frac{8t}{(t+1)^2} = \frac{\frac{8}{t}}{\left(1 + \frac{1}{t}\right)^2} \to 0 \qquad \text{as } t \to \infty$$

so both treatments predict that the concentration tends to zero. The difference between them is

$$\frac{8t}{(t+1)^2} - \frac{3(t+3)}{(t+1)^2} = \frac{5t - 9}{(t+1)^2}$$

which is positive for all $t > \frac{9}{5}$. So the second treatment, the tablet, gives the greater concentration for large values of t.